

Journal of Financial Crises

Volume 3 | Issue 2

2021

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Recommended Citation

Nunn, Sharon M. (2021) "Kyrgyz Republic's Debt Resolution Agency, DEBRA," *Journal of Financial Crises*: Vol. 3 : Iss. 2, 330-340.

DOI: <https://doi.org/10.17132/2693-3179.1187>

Available at: <https://elischolar.library.yale.edu/journal-of-financial-crises/vol3/iss2/17>

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Kyrgyz Republic's Debt Resolution Agency, DEBRA¹

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Yale Program on Financial Stability Case Study
June 23, 2021

Abstract

In the mid-1990s, the largest state-owned banks in the Kyrgyz Republic faced insolvency and a concomitant large stock of nonperforming loans, a problem stemming from the former Soviet Union's policy of directed credit to loss-making institutions. The government established DEBRA, a debt resolution agency and asset management company. DEBRA could liquidate or restructure a bank and take on its assets in the process, or just take on a bank's nonperforming assets. DEBRA received the assets in exchange for government securities. Staff attempted to resolve the debt by collection, restructuring, writing off, or liquidating the assets. Officials initially established DEBRA with a three-year mandate, but the entity eventually evolved into a semipermanent fixture in the country's financial system that still operated as of April 2021. By April 1999, the end of its initial mandate, DEBRA's portfolio of bad loans totaled Som 1,573 million (USD \$42 million), 73% of which were unresolved.

Keywords: Kyrgyz Republic, asset management companies, asset purchase programs, DEBRA, nonperforming loans, nonperforming assets

¹ This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering broad-based asset management company programs.

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Kyrgyz Republic's Debt Resolution Agency, DEBRA

At a Glance

Prior to the dissolution of the Soviet Union in 1991, the Kyrgyz Republic's financial system was structured to act as a regional component of the greater state system, which allocated resources according to the central government's expectations. Kyrgyz financial institutions lent to enterprises when directed to do so by Soviet officials and often did so without objective considerations of creditworthiness: the loans mostly covered losses or helped maintain employment. The Kyrgyz Republic government continued this practice until early 1994 (World Bank 1996).

After the dissolution of the Soviet Union in 1991, the Kyrgyz Republic experienced severe economic hardship caused by the cessation of former Soviet budget backing, previously amounting to approximately 10% of the country's gross domestic product (GDP), and trade. These repercussions continued to weigh on the Kyrgyz Republic in 1996. That year, the Kyrgyz Republic's financial system consisted of the country's central bank, then called the National Bank of Kyrgyzstan (NBK), 17 commercial banks, and a savings bank. Four of the banks, the so-called "Big Four" that were state-owned specialized banks, accounted for 80% of the system's assets (World Bank 1996). The 18 institutions' assets totaled 16.5% of the country's GDP in 1995, and only 8% of the system's gross loan portfolio was considered performing.⁴ Government-directed lending, as well as a culture of poor lending practices, led to the large stock of nonperforming loans. Additionally, the government offered the Big Four "implicit subsidies, including the free lease of many local premises and the

Summary of Key Terms	
Purpose: "acquire, collect or write-off the old non-performing loans" from the liquidation or restructuring of the "dominant and insolvent state banks" (World Bank 2000)	
Launch Dates	Formed: May 1996 Operational: October 1996
Wind-Down Dates	Initially three-year timeframe but extended; DEBRA's operations have not ceased as of April 2021
Size and Type of Nonperforming Loan Problem	92% of the financial system's loan portfolio was nonperforming in 1995 Primarily commercial loans
Program Size	Not specified at outset
Eligible Institutions	National Bank of Kyrgyzstan-approved banks; four state-owned banks initially targeted Open and closed-bank
Usage	Som 1,573 million (USD \$42 million) as of 1999 ³
Outcomes	Approx. 73% left to be resolved as of April 1999
Ownership Structure	Government-Owned
Notable Features	DEBRA continues to operate nearly 25 years after establishment

³ Documents detailing DEBRA programming did not include a US dollar equivalent for this figure. The author calculated the US dollar equivalent using the National Bank of Kazakhstan's historical exchange rate data available at <https://www.nbkr.kg/index1.jsp?item=2857&lang=ENG>.

⁴ The World Bank's 16.5% figure differs slightly from the Bloomberg data, 11.84%, shown in the Kyrgyz Republic context table below. A variety of technical explanations for this difference exist, such as subsequent data revisions related to definitional changes.

rescheduling of some NBK refinancing on soft terms" (World Bank 1996). Finally, poor lending practices in the immediate lead-up to the country's later financial system reforms occurred as the banks' shareholders and managers showed "little concern about further losses" because the institutions were already insolvent and "waiting for some form of a bail-out" from the government (World Bank 1996).

To stabilize the financial system, and by extension, help the broader economy, the Kyrgyz Republic undertook broad restructuring efforts, including updating the country's regulatory environment and dealing with the large insolvent state banks and their nonperforming loans. The latter part of the strategy began in earnest when the NBK board in 1996 created DEBRA, a debt restructuring agency and asset management company (AMC), to take control and dispose of assets and liabilities of banks the NBK sought to liquidate or restructure. The World Bank supported the government's efforts via a Financial Sector Adjustment Credit (FINSAC), a loan that helped fund DEBRA's operations and other financial system changes. The multilateral institution also facilitated the creation of the FINSAC Review Committee, which oversaw the implementation of reforms (World Bank 1996).

DEBRA could liquidate or restructure a bank and take on its assets in the process, or just take on a bank's nonperforming assets (World Bank 1996). Any institution or assets the NBK approved could be transferred to DEBRA, but officials initially and explicitly targeted the Big Four, liquidating two and restructuring two (World Bank 1996). During DEBRA's original three-year mandate, it also received nonperforming loans from eight other commercial banks and was appointed liquidator of an additional large bank (World Bank 2000). It is unclear if the eight commercial banks were restructured or liquidated.

DEBRA comprised two main units: the Bank Restructuring Unit (BRU), which performed the receiver and restructuring functions, and the Debt Recovery Unit (DRU), which dealt with the nonperforming loan portfolios transferred from the banks. Policymakers chose to transfer bad assets to the DRU "to avoid moral hazard and the loss of financial discipline that would result from wholesale, unconditional forgiveness of non-performing loans to the enterprise sector" (World Bank 1996).

The DRU recovered, sold, or wrote off the nonperforming assets it received. DRU staff had a variety of recovery and resolution methods at its disposal, including discounts, debt collateral swaps, debt restructuring, and the auction or sale of the debt (World Bank 1996). The agency originally planned to create automatic criteria for determining how to handle certain categories of loans but ended up creating individual plans for each debtor (World Bank 2000). The received assets were initially required to be resolved by the termination of DEBRA's original three-year mandate (World Bank 1996). Policymakers later extended the AMC's mandate by three additional years, to May 2002, to handle the liquidation of another large bank (World Bank 2000). Over subsequent years, DEBRA became a semipermanent, increasingly independent fixture in the Kyrgyz financial system, liquidating and rehabilitating banks and collecting debts on behalf of the Ministry of Finance (MOF) (World Bank 2015). DEBRA still operated as of April 2021.

During the three years of its initial mandate, DEBRA took on a total portfolio of Som 1,573 million (USD \$42 million), 73% of which was still unresolved as of April 1999 (World Bank 2000). DEBRA officials ascribed the poor resolution outcomes to problems with court-appointed executors and the particularly bad quality of the assets it handled (World Bank 2000). In addition to poor resolution results, the country's stock of nonperforming loans rose after DEBRA's intervention. Loans classified as "less than satisfactory" increased to 34% in 1999 from 9% in 1997 (World Bank 2000). It remains unclear if the nonperforming loans DEBRA originally received in the 1990s remain on the AMC's books today, but many of its liquidation cases had "been lingering for more than a decade," as of 2015 (World Bank 2015).

Summary Evaluation

Weak bank risk management, along with DEBRA's internal management structures and the 1998 Russian financial crisis, contributed to the AMC's poor resolution outcomes and the subsequent rise in bad assets in the Kyrgyz financial system, according to assessments by International Monetary Fund (IMF) and World Bank analysts. DEBRA's "technical and operational capacity," as well as its funding and legal status, impacted its ability to resolve the bad debt (IMF 2003). The AMC's eventual housing under the MOF hindered DEBRA's ability to "define the appropriate form of management, procedures, and guidelines to carry out its" functions, IMF analysts found (IMF 2003). In June 2002, DEBRA was granted legal status as the sole liquidator of failed banks, but because DEBRA was under the umbrella of the MOF, this led the MOF to have status as both the creditor and liquidator of a failed bank. Moreover, MOF and NBK transfers funded DEBRA, along with general operation fees. IMF analysts suggested that financing operations solely through fees might resolve conflicts of interest (IMF 2003).

Additionally, risk management in Kyrgyz Republic banks remained weak into the late 1990s, with "considerations other than risk management [guiding] the banks' lending decisions" in some instances (World Bank 2000).

Finally, Russia and its surrounding countries experienced a financial crisis in 1998, which contributed to the rise in bad assets in the Kyrgyz Republic (World Bank 2000). Some domestic banks "suffered some valuation losses from" the resultant depreciation of the som, "which had a significant impact on the quality of the banks' loan portfolios denominated in foreign currency" (Implementation Completion Report 2000). Additionally, Kyrgyz Republic exporters with markets based in Russia struggled to service their loans because of reduced demand (World Bank 2000).

The "Achilles heel of [the program] turned out to be the lack of adequate capacity and experience in banking supervision. Mostly this was a timing issue due to the fact that the financial crisis hit while [supervisory capacities were] being strengthened," World Bank researchers noted in a 2000 report (World Bank 2000).

Kyrgyz Republic's DEBRA: Kyrgyz Republic Context	
GDP (SAAR, nominal GDP in LCU converted to USD)	\$1.66 billion in 1995 \$1.83 billion in 1996
GDP per capita (SAAR, nominal GDP in LCU converted to USD)	\$364 in 1995 \$395 in 1996
Sovereign credit rating (five-year senior debt)	NA
Size of banking system	\$0.20 billion in 1995 \$0.17 billion in 1996
Size of banking system as a percentage of GDP	11.84% in 1995 9.41% in 1996
Size of banking system as a percentage of financial system	NA
Five-bank concentration of banking system	NA
Foreign involvement in banking system	NA
Government ownership of banking system	NA
Existence of deposit insurance	No
<i>Sources: Bloomberg; World Bank Global Financial Development Database; World Bank Deposit Insurance Dataset.</i>	

Key Design Decisions

- 1. Part of a Package: Kyrgyz Republic officials created DEBRA to acquire assets when liquidating or restructuring and recapitalizing financial institutions, in addition to taking on specific loans or assets of a bank identified by the relevant authorities.**

Policymakers created the AMC DEBRA to liquidate two dominant and insolvent state-owned banks and to restructure two others, while taking on their nonperforming loans where necessary. These banks, the so-called “Big Four,” accounted for 80% of the banking system’s assets and were a priority for officials looking to stabilize the financial system (World Bank 1996). Policymakers believed that the “strategy for these big banks [would] address the largest part of the bad asset problem in the banking sector” (World Bank 1996).

Two of the four, Agroprom Bank and Elbank, had nonperforming-to-gross-loan ratios of 98% and were slated for liquidation. AKB Kyrgyzstan Bank, with an 85% ratio, and Promstroi Bank, which had a 60% ratio, were downsized and restructured through private and government recapitalization. “The survival of the Big Four banks [had] been possible only because of specially negotiated terms for the NBK debt and because of exemption from prudential regulations” (World Bank 1996).

Lawmakers organized DEBRA into two main units to facilitate this strategy. Its Bank Restructuring Unit (BRU) performed a receivership function on behalf of the NBK, while its Debt Recovery Unit (DRU) managed and resolved the nonperforming loans transferred to it. The BRU assisted in the “intervention, receivership, conservatorship, deposit payout, and liquidation activities” of the banks and transferred nonperforming loans to the DRU (World Bank 1996). The DRU was set up to “act as the vehicle through which all recoveries [were] made by the Government” (World Bank 1996).

DEBRA could also take over other financial institutions and their assets (World Bank 2000).

- 2. Legal Authority: The NBK established DEBRA under the Resolution of the Board of NBKR No. 17/1.**

The resolution spelled out DEBRA’s restructuring, liquidating, and debt resolution functions in accordance with the Law of the Kyrgyz Republic “On bankruptcy (insolvency)” (World Bank 2000). The classification and restructuring processes for DEBRA were approved by the Resolution of the Government of the Kyrgyz Republic No. 412, dated August 28, 1996 (World Bank 2000). The FINSAC Review Committee, which approved the assets slated for transfer to DEBRA, was established on May 3, 1996, by a presidential decree (World Bank 1996).

- 3. Special Powers: DEBRA could pursue restitution from bank leaders and shareholders that contributed to the bankruptcy of their financial institutions (IMF 2003).**

The extent to which DEBRA officials have utilized this special power is unclear.

4. Mandate: Policymakers created DEBRA to resolve assets acquired from nonviable banks slated for liquidation and viable banks that were to be restructured.

Though the program initially targeted the Big Four, DEBRA could also take over other financial institutions and their assets, and it has done so.

5. Ownership Structure: DEBRA was government-owned.

DEBRA initially operated under the NBK (World Bank 2000). By 2003, the MOF owned DEBRA (IMF 2003). It had morphed into a “more independent entity reporting directly to the” country’s president by 2015 (World Bank 2015).

6. Governance/Administration: DEBRA was headed by a director who reported directly to the chairman of the NBK, and the FINSAC Review Committee approved assets slated for transfer to the AMC.

The director managed the day-to-day operations through two deputy directors. Though only the NBK chose which banks would be liquidated, and the NBK and MOF jointly determined assets to be transferred to DEBRA, the FINSAC Review Committee had to approve those transfers. The committee was also responsible for overseeing the implementation of government decisions stemming from the FINSAC program that the World Bank sponsored. The FINSAC Review Committee was chaired by the prime minister or his designee, while its membership was composed of DEBRA’s director; the chairmen of the NBK and the State Property Fund; and the ministers of finance, economy, agriculture, and justice (World Bank 1996).

DEBRA’s BRU comprised primarily NBK liquidation personnel. NBK and MOF employees staffed the DRU. The BRU received two foreign advisers “experienced in debt recovery methodologies, receivership procedures, and bank liquidations,” while the DRU received two foreign experts with experience in debt workouts and recovery (World Bank 1996).

DEBRA was required to submit periodic reports to the NBK’s board and the Kyrgyz government and to cooperate with any mandatory audits (World Bank 1996).

It is unclear how DEBRA’s governance and staffing structure changed as the organization’s mandate was extended and then exceeded.

7. Size: Kyrgyz Republic officials do not appear to have established a size cap for DEBRA.

World Bank documents detailing the program did not reference a size limit, and, in fact, granted that banks other than the Big Four could be liquidated and/or restructured and have their assets transferred to DEBRA (World Bank 1996). During the three years of its initial mandate, DEBRA took on debt worth Som 1,573 million (USD \$42 million) (World Bank 2000).

8. Funding Sources: The Kyrgyz Republic government and the World Bank funded DEBRA’s asset acquisitions.

World Bank analysts projected that resources needed for DEBRA’s first year of operation would total 300-350 million Soms (a maximum of USD \$32 million) or about 1.5% of GDP. “Thereafter, the program would cost between 15-59 million Soms per annum, mainly as

interest on bonds to be issued as part of the bank restructuring program” (World Bank 1996).

The World Bank’s FINSAC loan, which totaled USD \$45 million, was earmarked to pay for the first year’s costs and an additional USD \$13 million in budget support during the rest of DEBRA’s initial mandate (World Bank 1996). The loan was disbursed in two tranches: the first in September 1996, and the second in November 1997 after the government met certain conditions of the agreement (World Bank 2000). Other costs of the banking system restructuring were financed by the national budget, “after first imposing losses on the banks’ shareholders” (World Bank 1996). These costs included “payments to depositors, the removal of bad assets from bank balance sheets,” and severance payments, among other outlays (World Bank 1996).

Policymakers assumed at the outset that “collection and liquidation costs [would] be covered by the collections made against bad assets and the sale of assets under liquidation” (World Bank 1996). A follow-up report stated that DEBRA was eventually self-financing, funded by “an allocation of 5 percent of the debts it recovers” (World Bank 2000).

By 2003, however, DEBRA operated off of a combination of fees and transfers from the MOF and NBK (IMF 2003).

9. Eligible Institutions: Policymakers created DEBRA chiefly to liquidate and restructure the Big Four, but officials could choose to liquidate or restructure other banks as well with NBK approval.

The NBK had the sole authority to choose banks other than the Big Four to liquidate, and the NBK and MOF jointly recommended assets to transfer to DEBRA. The FINSAC Review Committee was responsible for approving the asset transfers. Though the actual eligibility criteria for transferrable institutions is unclear within the available documentation, state-owned banks were a “top priority” for the country’s restructuring strategy (World Bank 1996).

10. Eligible Assets: DEBRA took assets from restructured and liquidated banks and could also receive assets from financial institutions that DEBRA did not restructure or liquidate.

The NBK and MOF jointly recommended assets to transfer to DEBRA, and the FINSAC Review Committee approved the asset transfers (World Bank 1996). It is unclear within available documentation how the assets were chosen.

11. Acquisition (Mechanics): DEBRA received the banks’ nonperforming assets in exchange for government securities.

The type of government security or securities used is unclear. The Kyrgyz Republic was in the process of developing its government securities market after DEBRA was created, preparing regulations in 2000 to issue government medium-term bonds with maturities from two to five years (World Bank 2000).

12. Acquisition (Pricing): It is unclear how DEBRA officials priced the assets the AMC received.

Available program documents do not contain information about DEBRA's pricing methodology.

13. Recovery and Disposal: The DRU's primary debt collection strategy was to utilize the most rapid and economical methods when trying to resolve bad assets.

The BRU created "detailed" liquidation programs for Agroprom Bank and Elbank, which were both placed under the receivership of DEBRA to start the process (World Bank 2000). The strategy included selling or closing bank branches and paying off small depositors while transferring larger ones to a dedicated government entity. Diagnostic audits were prepared for Promstroi Bank and AKB Kyrgyzstan Bank, "and an internally managed program was implemented under a Management Letter Agreement with NBK, which provided for strong loan collection efforts, write offs and adequate loan-loss provisions, as well as downsizing and rationalization measures" (World Bank 2000).

Documents detailing DEBRA's original design suggested that the AMC would develop automatic criteria that would streamline debt recovery actions where possible (World Bank 1996). However, DEBRA officials created individual action plans for each debtor, a strategy "that did not fully meet the initial understanding" of the World Bank, which helped fund the restructuring project (World Bank 2000). At any rate, DRU staff could use a variety of recovery methods, split into four categories detailed in Figure 1 below.

Figure 1: DEBRA Recovery Methods

Method	Description
Restructuring	Placing the loan on a performing basis
Forgiveness	Partial forgiveness of either interest and/or loan principal and the restructuring of the remaining loan balance
Liquidation	Liquidating through collection efforts and/or sale of collateral; or selling through the sealed-bid process, auctions, or packaged sale mechanism to interested buyers or investors
Write-Off	Writing off all uncollectable principal and interest

Source: World Bank 1996.

The government also introduced the leasing auction, which was then a new asset disposal method for the country. Details on how this methodology worked in practice, and whether or how often it was used, are unclear from available documentation. Additionally, in remote areas of the Kyrgyz Republic, "agricultural products were exchanged for debts" (World Bank 2000).

14. Timeframe: Officials originally mandated that DEBRA operate for three years. The government then extended the mandate an additional three years before DEBRA later became a semipermanent fixture in the Kyrgyz Republic financial system.

The NBK formed DEBRA in May 1996, and it became fully operational in October that year (World Bank 2000). DEBRA's initial mandate was scheduled to terminate in Spring 1999. All DRU assets were "to be recovered, sold, or written off and the proceeds remitted to the central Government budget authorities by the termination date of DEBRA's mandate" (World Bank 1996). Similarly, all receivership assets were "to be liquidated and the proceeds remitted to the proper authorities by the termination date of DEBRA's mandate" (World Bank 1996).

Policymakers extended DEBRA's mandate to May 2002 after the AMC was appointed liquidator of another bank (World Bank 2000). DEBRA was later granted the status of "sole liquidator of failed banks" in a revised bankruptcy law that was enacted in June 2002 (IMF 2003). It "subsequently evolved into an asset management company with multiple tasks, and became a more independent entity reporting directly to the President," with these changes formalized in August 2011 (World Bank 2015). Various attempts have been made to reform or close units of DEBRA in the past decade (World Bank 2015). For example, the country hired Deloitte to audit DEBRA with the goal of winding down DEBRA's assets (Lavrova and Jenish 2015).

Still, Kyrgyz Republic politicians adjusted DEBRA's functions in 2017 and 2018. DEBRA operated and was in charge of liquidating three bankrupt banks as of April 2021.

References and Key Program Documents

Implementation Documents

(Lavrova and Jenish 2015) Lavrova, Olga, and Nurbek Jenish. 2015. "Kyrgyz Republic: Letter of Intent, Memorandum of Economic and Financial Policies, and Technical Memorandum of Understanding." *Letter and memoranda from the Kyrgyz Republic government describing policies the country intended to implement in the context of its request for financial support from the IMF*. <https://ypfs.som.yale.edu/library/kyrgyz-republic-letter-intent-memorandum-economic-and-financial-policies-and-technical>.

(World Bank 1996) World Bank. 1996. "Report and Recommendation of the President of the International Development Association to the Executive Directors on a Proposed Financial Sector Adjustment Credit in an Amount Equivalent to SDR 31.2 Million to the Kyrgyz Republic." World Bank Report No. P-6891-KG. May 31, 1996. *World Bank report outlining proposed funding for a financial sector reform project in the Kyrgyz Republic*. <https://ypfs.som.yale.edu/library/report-and-recommendation-president-international-development-association-executive>.

(World Bank 2000) World Bank. 2000. "Implementation Completion Report: Kyrgyz Republic Financial Sector Adjustment Credit (FINSAC)." World Bank Report No. 19352. June 30, 2000. *World Bank report summarizing the results of Kyrgyz Republic financial sector reforms the multilateral organization helped fund*.

<https://ypfs.som.yale.edu/library/implementation-completion-report-kyrgyz-republic-financial-sector-adjustment-credit-finsac>.

Reports/Assessments

(IMF 2003) International Monetary Fund (IMF). 2003. "Kyrgyz Republic: Financial System Stability Assessment, Including Reports on the Observance of Standards and Codes on the Following Topics: Monetary and Financial Policy Transparency, Banking Supervision, and Payment Systems." IMF Country Report No. 03/52. February 19, 2003.

IMF staff country report assessing the Kyrgyz Republic's financial system.

<https://ypfs.som.yale.edu/library/kyrgyz-republic-financial-system-stability-assessment-including-reports-observance>.

(World Bank 2015) World Bank. 2015. "Project Information Document (PID) Identification/Concept Stage: Audit of DEBRA." World Bank Report No. PIDC52427. December 15, 2015.

Report proposing an audit of Kyrgyz Republic asset management company DEBRA.

<https://ypfs.som.yale.edu/library/project-information-document-pid-identificationconcept-stage-audit-debra>.

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